

# Solar C appendix

Sun · C · downloadable appendix

Solar carbon:  $8.430 \pm 0.072$

Asplund, Amarsi & Grevesse 2021:  $8.46 \pm 0.04$  · this measurement -0.030 dex

Kitt Peak · VIS · Reference Grade · SET-AGSS21 · ENGINE-A-3DNLTE ·  $n = 3$ . The top-ranked admitted product (Reference Grade, then the most complete treatment, then line count, then total uncertainty); independent products and engines are not averaged together.

## Highlighted Products

### VIS

**$8.430 \pm 0.072$**

$\pm 0.052$  stat ·  $\pm 0.041$  syst

SET-AGSS21 · C I · Synth · 3D-NLTE · Amarsi · SYNTH · ENGINE-A-3DNLTE · line set not supplied · solar\_kpno\_molecfits\_corrected · Reference Grade

### red-optical

**$8.488 \pm 0.069$**

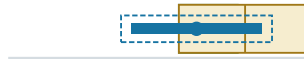
$\pm 0.036$  stat ·  $\pm 0.050$  syst

SET-AGSS21 · C I · Synth · 3D-NLTE · Amarsi · SYNTH · ENGINE-A-3DNLTE · line set not supplied · solar\_iag · Reference Grade

## Error-bar Forest Plot

### VIS — harps\_solar\_harps\_molecfits\_corrected A(C) dex; axis 8.19-8.78

ATOM-CI\_5052 · C I · Synth · 1D-LTE · SYNTH ·  
1D-LTE · line set not supplied  
Codex Grade · ALL · n=1



8.444  
±0.089 stat ±0.103 syst

ATOM-CI\_5380 · C I · Synth · 1D-LTE · SYNTH ·  
1D-LTE · line set not supplied  
Codex Grade · ALL · n=1



8.528  
±0.087 stat ±0.218 syst

MOL-C2\_Swan · C I · Synth · 1D-LTE · SYNTH · 1D-LTE  
· line set not supplied  
Codex Grade · ALL · n=167



8.419  
±0.010 stat ±0.186 syst

MOL-CH\_Gband · C I · Synth · 1D-LTE · SYNTH ·  
1D-LTE · line set not supplied  
Codex Grade · ALL · n=184



8.509  
±0.006 stat ±0.089 syst

SET-AGSS21 · C I · Synth · 1D-LTE · SYNTH · 1D-LTE ·  
line set not supplied  
Reference Grade · ALL · n=3



8.483  
±0.019 stat ±0.041 syst

SET-AGSS21 · C I · Synth · 3D-NLTE · Amarsi · SYNTH  
· ENGINE-A-3DNLTE · line set not supplied  
Reference Grade · ALL · n=3



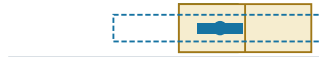
8.463  
±0.018 stat ±0.041 syst

SET-AGSS21 · C I · Synth · 1D-NLTE · Bergemann ·  
SYNTH · ENGINE-A · line set not supplied  
Reference Grade · ALL · n=3



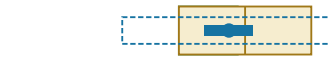
8.476  
±0.019 stat ±0.041 syst

indicator not supplied · C I · Synth · 3D-NLTE ·  
Amarsi · SYNTH · ENGINE-A-3DNLTE · line set not  
supplied  
Codex Grade · ALL · n=2



8.476  
±0.031 stat ±0.145 syst

indicator not supplied · C I · Synth · 1D-NLTE ·  
Bergemann · SYNTH · ENGINE-A · line set not  
supplied  
Codex Grade · ALL · n=2



8.488  
±0.033 stat ±0.145 syst

### VIS — iag\_fts\_solar\_atlas solar\_iag A(C) dex; axis 8.19-8.78

indicator not supplied · C I · Synth · 1D-LTE · SYNTH ·  
1D-LTE · line set not supplied  
Codex Grade · ALL · n=2



8.373  
±0.114 stat ±0.041 syst

indicator not supplied · C I · Synth · 3D-NLTE ·  
Amarsi · SYNTH · ENGINE-A-3DNLTE · line set not  
supplied  
Codex Grade · ALL · n=2



8.355  
±0.115 stat ±0.041 syst

indicator not supplied · C I · Synth · 1D-NLTE ·  
Bergemann · SYNTH · ENGINE-A · line set not  
supplied  
Codex Grade · ALL · n=2



8.367  
±0.113 stat ±0.041 syst

### VIS — kpno\_solar\_atlas solar\_kpno\_kurucz2005\_corrected A(C) dex; axis 8.19-8.78

SET-AGSS21 · C I · Synth · 1D-LTE · SYNTH · 1D-LTE ·  
line set not supplied  
Reference Grade · ALL · n=3



8.467  
±0.059 stat ±0.041 syst

## VIS — kpno\_solar\_atlas solar\_kpno\_kurucz2005\_corrected A(C) dex; axis 8.19-8.78

SET-AGSS21 · C I · Synth · 3D-NLTE · Amarsi · SYNTH  
· ENGINE-A-3DNLTE · line set not supplied  
Reference Grade · ALL · n=3



8.449  
±0.059 stat ±0.041 syst

SET-AGSS21 · C I · Synth · 1D-NLTE · Bergemann ·  
SYNTH · ENGINE-A · line set not supplied  
Reference Grade · ALL · n=3



8.460  
±0.059 stat ±0.041 syst

indicator not supplied · C I · Synth · 3D-NLTE ·  
Amarsi · SYNTH · ENGINE-A-3DNLTE · line set not  
supplied  
Codex Grade · ALL · n=2



8.363  
±0.086 stat ±0.145 syst

indicator not supplied · C I · Synth · 1D-NLTE ·  
Bergemann · SYNTH · ENGINE-A · line set not  
supplied  
Codex Grade · ALL · n=2



8.375  
±0.085 stat ±0.145 syst

## VIS — kpno\_solar\_atlas solar\_kpno\_molecfit\_corrected A(C) dex; axis 8.19-8.78

SET-AGSS21 · C I · Synth · 1D-LTE · SYNTH · 1D-LTE ·  
line set not supplied  
Reference Grade · ALL · n=3



8.450  
±0.052 stat ±0.041 syst

SET-AGSS21 · C I · Synth · 3D-NLTE · Amarsi · SYNTH  
· ENGINE-A-3DNLTE · line set not supplied  
Reference Grade · ALL · n=3



8.430  
±0.052 stat ±0.041 syst

SET-AGSS21 · C I · Synth · 1D-NLTE · Bergemann ·  
SYNTH · ENGINE-A · line set not supplied  
Reference Grade · ALL · n=3



8.443  
±0.052 stat ±0.041 syst

indicator not supplied · C I · Synth · 3D-NLTE ·  
Amarsi · SYNTH · ENGINE-A-3DNLTE · line set not  
supplied  
Codex Grade · ALL · n=2



8.376  
±0.085 stat ±0.145 syst

indicator not supplied · C I · Synth · 1D-NLTE ·  
Bergemann · SYNTH · ENGINE-A · line set not  
supplied  
Codex Grade · ALL · n=2



8.388  
±0.084 stat ±0.145 syst

VIS axis: 8.19 · 8.33 · 8.48 · 8.63 · 8.78 A(C) dex

## red-optical — iag\_fts\_solar\_atlas solar\_iag A(C) dex; axis 8.39-8.70

SET-AGSS21 · C I · Synth · 1D-LTE · SYNTH · 1D-LTE ·  
line set not supplied  
Reference Grade · ALL · n=10



8.529  
±0.039 stat ±0.050 syst

SET-AGSS21 · C I · Synth · 3D-NLTE · Amarsi · SYNTH  
· ENGINE-A-3DNLTE · line set not supplied  
Reference Grade · ALL · n=10



8.488  
±0.036 stat ±0.050 syst

SET-AGSS21 · C I · Synth · 1D-NLTE · Bergemann ·  
SYNTH · ENGINE-A · line set not supplied  
Reference Grade · ALL · n=10



8.490  
±0.035 stat ±0.050 syst

indicator not supplied · C I · Synth · 1D-LTE · SYNTH ·  
1D-LTE · line set not supplied  
Codex Grade · ALL · n=6



8.566  
±0.037 stat ±0.050 syst

indicator not supplied · C I · Synth · 3D-NLTE ·  
Amarsi · SYNTH · ENGINE-A-3DNLTE · line set not  
supplied  
Codex Grade · ALL · n=6



8.528  
±0.040 stat ±0.050 syst

## red-optical — iag\_fts\_solar\_atlas solar\_iag A(C) dex; axis 8.39-8.70

indicator not supplied · C I · Synth · 1D-NLTE ·  
Bergemann · SYNTH · ENGINE-A · line set not  
supplied  
Codex Grade · ALL · n=6



8.527  
±0.039 stat ±0.050 syst

## red-optical — kpno\_solar\_atlas solar\_kpno\_kurucz2005\_corrected A(C) dex; axis 8.39-8.70

SET-AGSS21 · C I · Synth · 3D-NLTE · Amarsi · SYNTH  
· ENGINE-A-3DNLTE · line set not supplied  
Reference Grade · ALL · n=10



8.536  
±0.042 stat ±0.050 syst

SET-AGSS21 · C I · Synth · 1D-NLTE · Bergemann ·  
SYNTH · ENGINE-A · line set not supplied  
Reference Grade · ALL · n=10



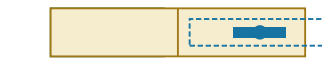
8.532  
±0.041 stat ±0.050 syst

indicator not supplied · C I · Synth · 1D-LTE · SYNTH ·  
1D-LTE · line set not supplied  
Codex Grade · ALL · n=6



8.610  
±0.018 stat ±0.050 syst

indicator not supplied · C I · Synth · 3D-NLTE ·  
Amarsi · SYNTH · ENGINE-A-3DNLTE · line set not  
supplied  
Codex Grade · ALL · n=6



8.568  
±0.019 stat ±0.050 syst

indicator not supplied · C I · Synth · 1D-NLTE ·  
Bergemann · SYNTH · ENGINE-A · line set not  
supplied  
Codex Grade · ALL · n=6



8.563  
±0.018 stat ±0.050 syst

## red-optical — kpno\_solar\_atlas solar\_kpno\_molecfit\_corrected A(C) dex; axis 8.39-8.70

SET-AGSS21 · C I · Synth · 1D-LTE · SYNTH · 1D-LTE ·  
line set not supplied  
Reference Grade · ALL · n=10



8.576  
±0.047 stat ±0.050 syst

SET-AGSS21 · C I · Synth · 3D-NLTE · Amarsi · SYNTH  
· ENGINE-A-3DNLTE · line set not supplied  
Reference Grade · ALL · n=10



8.536  
±0.045 stat ±0.050 syst

SET-AGSS21 · C I · Synth · 1D-NLTE · Bergemann ·  
SYNTH · ENGINE-A · line set not supplied  
Reference Grade · ALL · n=10



8.537  
±0.043 stat ±0.050 syst

indicator not supplied · C I · Synth · 1D-LTE · SYNTH ·  
1D-LTE · line set not supplied  
Codex Grade · ALL · n=6



8.616  
±0.020 stat ±0.050 syst

indicator not supplied · C I · Synth · 3D-NLTE ·  
Amarsi · SYNTH · ENGINE-A-3DNLTE · line set not  
supplied  
Codex Grade · ALL · n=6



8.579  
±0.022 stat ±0.050 syst

indicator not supplied · C I · Synth · 1D-NLTE ·  
Bergemann · SYNTH · ENGINE-A · line set not  
supplied  
Codex Grade · ALL · n=6



8.582  
±0.022 stat ±0.050 syst

red-optical axis: 8.39 · 8.47 · 8.54 · 8.62 · 8.70 A(C) dex

**How to read this forest.** Each row is one independent product, preserving its band, holding, engine, indicator and line set. Engines and indicator families are never averaged. Solid bars show the supplied statistical uncertainty; wireframes show the supplied systematic uncertainty. The statistical basis and any incomplete error budget remain attached to the row. Grades describe the supplied line-pool provenance, not engine quality; evaluated theoretical CNO oscillator strengths are not relabelled as laboratory evidence.

Quarantined, failed and unconstrained fits have no abundance mark. Experimental products remain explicitly not adopted. Literature references provide context, never tuning targets.

## How this was measured

Every carbon value on this page comes from fitting synthetic spectra to the observed solar spectrum, line by line or band by band, with the carbon abundance as the free parameter. The synthesis includes the atomic lines and the molecular bands (CN, CH, C<sub>2</sub>, OH and others) in every window, so blended absorption is modelled rather than attributed to the line being measured. The continuum is placed locally around every line by comparing the observation with the synthesis on its own highest pixels, and ground-based spectra are telluric-corrected across their full range.

Each indicator is measured independently on each solar spectrum and reported separately; nothing is averaged across indicators or instruments. Where a 3D or NLTE correction is published for the lines, it is applied as its own product (1D-LTE, 1D-NLTE and 3D-NLTE are separate rows). Every value carries a full uncertainty budget: fit, line strengths, microturbulence, continuum, fit window, model atmosphere, telluric residual, blends and coupling to the other C/N/O abundances, and the spread between spectra, each measured by refitting with that one input changed. The headline at the top is the Asplund-grade product: a published line set, the most complete treatment, then the most lines.

## Solar carbon context

Carbon is traced by atomic C I, including forbidden [C I], and molecular CH, C<sub>2</sub> and CO indicators. Each probes different line formation and molecular-equilibrium sensitivities. Indicator families and atmospheric treatments remain separate products.

[C/O indicator strategy](#) · [Methodology](#)

## Download the evidence

[Download PDF — Solar C appendix](#)

[Source feed — includes quarantined fit outputs, not adopted abundances](#)

[Complete product visibility audit](#)

[Shared website/PDF report](#)

[Reproducibility manifest](#)

[Complete references and product-to-source bindings](#)

## Reproducibility

Generated from the pinned CNO campaign snapshot. Source commit

5fee27a14f2ab28d433b5f0b20101ce24f5118e1

Feed C.json v1.163 · updated 2026-10-03T22:11:25Z

Generated 2026-10-03T16:12:36-06:00 (source commit timestamp)

Feed SHA-256 f3146e51b5f7493e4bcec5843bb4cfbfab7995f56be16c85668a009ed80c23b5

Generator scripts/generate\_cno\_publication.py v1.0.0

[Pinned scientific source](#)

## References / Data & Model Sources

Sources are separated by their role. Registered grid lineage and quarantined diagnostics do not imply an adopted abundance or an applied correction.

See the source notes for the distinction between reference lineage and the treatment actually recorded in each feed entry.

[Science and provenance source notes](#)

**Solar abundance and indicator analyses.** Solar C atomic/molecular indicator lineage and atmospheric sensitivity; comparison literature, not an adopted Codex abundance. Sources: [1], [2].

**Atomic data and gf authorities.** The recorded C I/II pedigree is the partial NIST MCHF update described by Wiese & Fuhr; evaluated theory is not primary laboratory evidence. Sources: [3], [4].

**Atomic data and gf authorities.** AGSS21 names Li et al. for its C I gf lineage. This reference calculation is distinct from the campaign's NIST/MCHF adjudication; no gf source is substituted here. Sources: [5].

**Model and grid sources.** Registered C I line/grid lineage: Amarsi, Nissen & Skuladottir 2019, A&A 630, A104 — line-by-line 3D-NLTE / 1D-NLTE grids (CDS J/A+A/630/A104). The current feed treatment remains authoritative; a grid reference does not imply a correction was applied. Sources: [6].

**Atomic data and gf authorities.** Atomic C I entry; source-family pedigree only, not a substitute for a per-transition gf join. Sources: [3], [4].

**Model and grid sources.** Feed treatment: 1D-LTE; no NLTE/3D correction is claimed for this entry. Turbospectrum is the synthesis implementation. Sources: [7].

**Molecular line data and delivery.** C2 line-data delivery: C2 Swan  $d^3\Pi_g - a^3\Pi_u$  (Brooke et al. 2013 / Masseron compilation). Masseron compilation, VALD-based, distributed with iSpec/Turbospectrum (Gerber et al. 2023, A&A 669, A43; uploaded 2022 to keeper.mpd.l.mpg.de, split per wavelength region) — see molecules-dir README.md Sources: [2], [7], [8].

**Molecular line data and delivery.** CH line-data delivery: CH A-X: Masseron et al. 2014 (MNRAS 442, 1489) / 2022 update. Masseron compilation, VALD-based, distributed with iSpec/Turbospectrum (Gerber et al. 2023, A&A 669, A43; uploaded 2022 to keeper.mpd.l.mpg.de, split per wavelength region) — see molecules-dir README.md The primary intake identifies Masseron 2014 as A&A 571 A47; the older delivery manifest's MNRAS locator conflicts with that record. Sources: [2], [7], [9].

**Model and grid sources.** Registered C/O model lineage; per-product grid provenance must confirm the applied treatment. Run provenance: Amarsi, Nissen & Skuladottir 2019, A&A 630, A104 — 3D non-LTE - 1D LTE on STAGGER (table2 C I / table3 O I); PER-LINE additive correction, grid lines only (match  $\leq 0.05$  Å) Sources: [6].

**Model and grid sources.** Registered C/O model lineage; per-product grid provenance must confirm the applied treatment. Run provenance: Amarsi, Nissen & Skuladottir 2019, A&A 630, A104 — 1D non-LTE - 1D LTE (table5 C I / table6 O I); PER-LINE additive correction, grid lines only (match  $\leq 0.05$  Å) Sources: [6].

Product-to-source index

C-products-45eae148b981 · products · VIS · solar\_harps\_molecfit\_corrected · ATOM-CI\_5052 · 1D-LTE · not supplied · [1], [3], [4], [7]

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C-products-e70e17e756dc · products · red-optical · solar\_kpno\_molecfit\_corrected · not supplied · ENGINE-A · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[6\]](#)

C-quarantine-450b744f3f97 · quarantine · VIS · solar\_harps\_molecfit\_corrected · not supplied · 1D-LTE · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[7\]](#)

C-quarantine-2f69a108b1e7 · quarantine · VIS · solar\_kpno\_kurucz2005\_corrected · not supplied · 1D-LTE · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[7\]](#)

C-quarantine-4e2a7411ee02 · quarantine · VIS · solar\_kpno\_molecfit\_corrected · not supplied · 1D-LTE · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[7\]](#)

C-quarantine-09ea5d36f2c8 · quarantine · red-optical · solar\_kpno\_kurucz2005\_corrected · SET-AGSS21 · 1D-LTE · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[7\]](#)

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